

Lecture 13

Bias–variance decomposition, MAP

COURSE MATH 637: Mathematical Techniques in Data Science

WEBSITE <https://jacobcalvert.com/teaching/math637fall2026/>

INSTRUCTOR Jacob Calvert (calvert@udel.edu)

OFFICE HOURS MW 2:45pm–3:45pm, Ewing Hall 509

Announcements & Reminders

- Quiz 3 on linear regression will be Friday.
- I plan to post grades for HW1 and HW2 on Friday.

Plan for today

- | | |
|---|--------|
| • Bias–variance decomposition | 20 min |
| • Ridge accepts bias to reduce variance | 7 min |
| • From maximum likelihood to MAP | 13 min |
| • MAP with a Gaussian prior | 10 min |

Why does a response contain **noise**?

Example (Advertising data)

- two markets with the same ad budgets need not have the same sales

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Noise is any variation in response y_i that the features x_i do not explain.

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Idea: Model *average* response to input x_i by $f(x_i)$ and noise by ε_i :

$$y_i = f(x_i) + \varepsilon_i.$$

By definition, ε_i has mean zero.

One dataset, many **possible** datasets

Example (Advertising data, continued)

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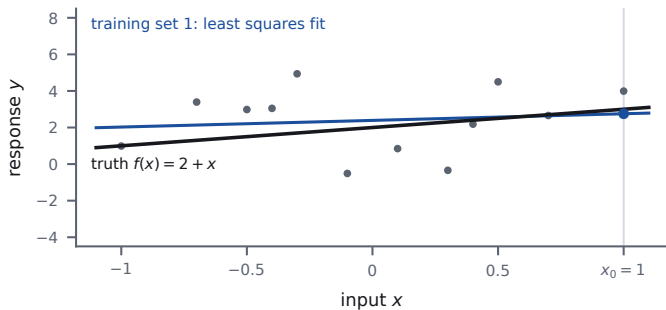
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Idea: We evaluate a fitting procedure by its performance across the datasets this collection process *could have produced*.

Example: Different training sets and fits

Held fixed: inputs $x_1, \dots, x_{12}$ and average response $f(x) = 2 + x$.

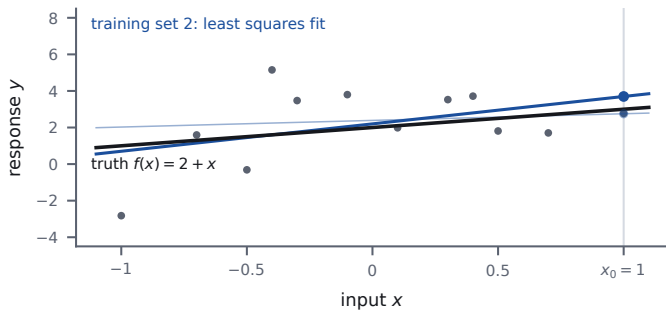
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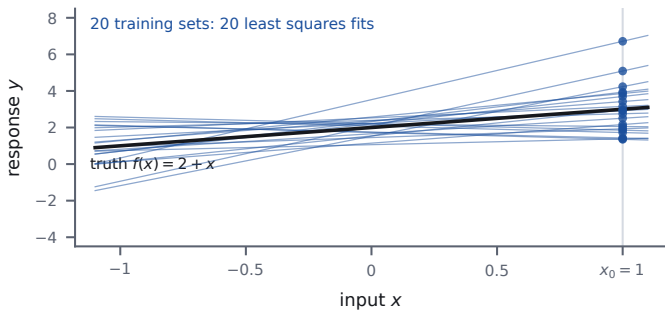
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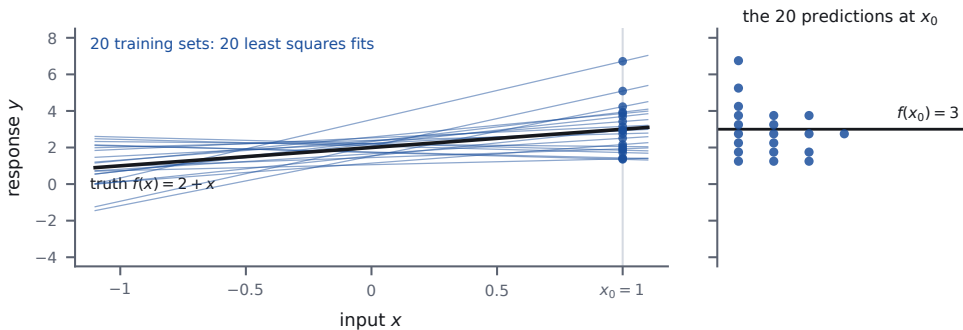
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Each fitted line $\hat{f}_D$ gives one prediction at $x_0 = 1$: the 20 dots on the right.

Evaluating the procedure that fits $\hat{f}_D$

Idea: We evaluate a fitting procedure by its performance across the datasets this collection process *could have produced*.

Given fitted $\hat{f}_D$, consider predicting a new response at input x_0 :

$$\underbrace{y_0 = f(x_0) + \varepsilon_0}_{\text{new response}}$$

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The loss for a fixed dataset D and ε_0 is the squared error

$$(y_0 - \hat{f}_D(x_0))^2.$$

We evaluate the procedure that fits $\hat{f}_D$ by averaging over D and ε_0 :

$$\text{MSE}(x_0) = \mathbb{E}_{D, \varepsilon_0} \left[(y_0 - \hat{f}_D(x_0))^2 \right].$$

Definition: Bias and variance

Given fitted $\hat{f}_D$, consider predicting a new response at input x_0 :

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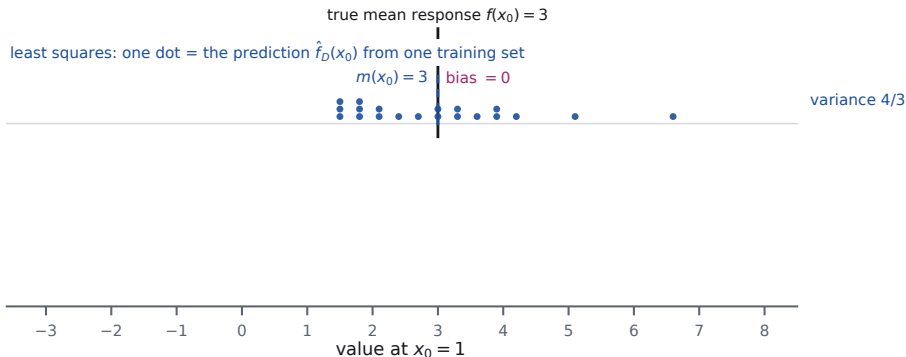
The *bias* is how much $m(x_0)$ differs from the true average response $f(x_0)$:

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The *variance* of the prediction at x_0 around $m(x_0)$ is

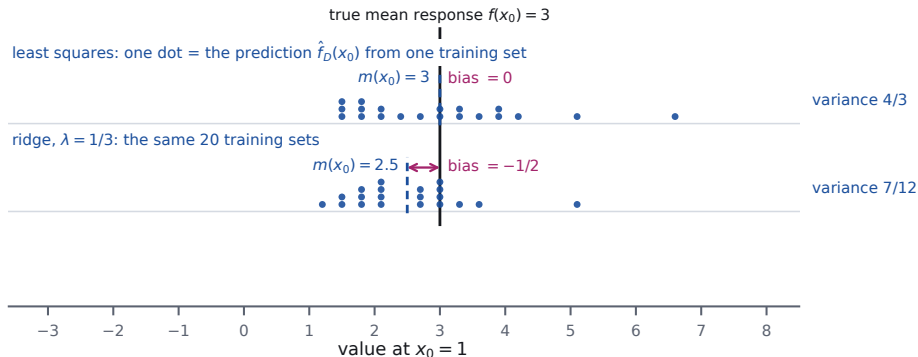
$$\text{var}(x_0) = \mathbb{E}_D \left[(\hat{f}_D(x_0) - m(x_0))^2 \right].$$

Example: Predictions at x_0 on one axis



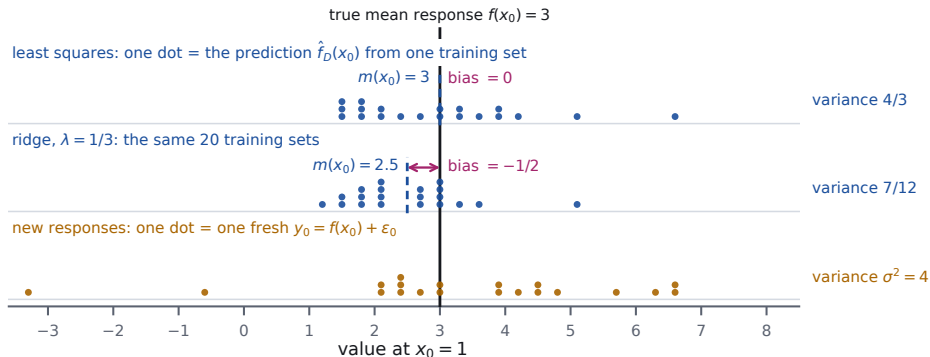
Here f is known, so $m(x_0)$ can be computed exactly. Least squares has $m(x_0) = f(x_0) = 3$: bias 0. The spread of the dots illustrates the variance.

Example: Predictions at x_0 on one axis



Ridge, with $\lambda = 1/3$ held fixed: the predictions are centered at $m(x_0) = 2.5$. The gap between the two centers is the bias; it can have either sign.

Example: Predictions at x_0 on one axis



A new response is centered at $f(x_0)$, not at $m(x_0)$. Its spread is the same whatever we predict.

The noise in a new response

Even if we always predicted the true mean response, a new outcome could differ from it:

$$\mathbb{E}_{\varepsilon_0}[(y_0 - f(x_0))^2] = \mathbb{E}[\varepsilon_0^2] = \sigma^2.$$

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Note: Training noise makes the fitted models vary. The new response's noise is a separate contribution to the prediction error.

Theorem: Bias–variance decomposition

Idea: We evaluate a fitting procedure by its performance across the datasets this collection process *could have produced*.

Theorem

Assuming the noise ε_0 has variance σ^2 and is independent of D ,

$$\text{MSE}(x_0) = \underbrace{(m(x_0) - f(x_0))^2}_{\text{bias}(x_0)^2} + \underbrace{\text{Var}_D[\hat{f}_D(x_0)]}_{\text{var}(x_0)} + \underbrace{\sigma^2}_{\text{noise strength}}.$$

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Note: f need not be linear; the noise need not be Gaussian.

Proof: Why the three terms add

Write $f_0 = f(x_0)$, $\hat{f}_0 = \hat{f}_D(x_0)$, and $m_0 = \mathbb{E}[\hat{f}_0]$, and drop the D, ε_0 subscripts.

- **Step 1.** Separate new response noise from error in estimating f_0 :

$$\text{MSE}(x_0) = \mathbb{E}[(y_0 - \hat{f}_0)^2] = \mathbb{E}[(\varepsilon_0 + f_0 - \hat{f}_0)^2] = \sigma^2 + \mathbb{E}[(\hat{f}_0 - f_0)^2].$$

Holds because $\mathbb{E}[\varepsilon_0] = 0$ and ε_0 is independent of $\hat{f}_0$.

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- **Step 2.** Separate bias from the variance of $\hat{f}_0$:

$$\mathbb{E}[(\hat{f}_0 - f_0)^2] = \mathbb{E}[((\hat{f}_0 - m_0) + (m_0 - f_0))^2] = \text{Var}(\hat{f}_0) + (m_0 - f_0)^2.$$

Holds because $\mathbb{E}[\hat{f}_0 - m_0] = 0$ and $m_0 - f_0$ is a constant.

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Observation: Least squares is unbiased

Suppose the average response $f(x)$ is *actually* linear, meaning there is some θ such that $f(x) = \theta^\top x$.

Claim

Averaging the least squares prediction $\hat{f}_D(x) = \hat{\theta}_D^\top x$ over datasets D gives $f(x)$, i.e., $\text{bias}(x) = 0$ for every x .

- Since θ is the “true” parameter vector, $y = X\theta + \varepsilon$ where $\mathbb{E}[\varepsilon] = 0$.

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- This implies unbiasedness because

$$\text{bias}(x) = m(x) - f(x) = \mathbb{E}[\hat{\theta}_D^\top x] - \theta^\top x = \mathbb{E}[\hat{\theta}_D]^\top x - \theta^\top x = 0.$$

Key idea: “Shrink” $\hat{\theta}$

For simplicity, suppose we fit **one coefficient**, $\hat{\theta}_{\text{LS}}$, using least squares.

Multiply $\hat{\theta}_{\text{LS}}$ by $a \in [0, 1]$ to get $\hat{\theta}_a = a \hat{\theta}_{\text{LS}}$. What happens?

- Denoting $\text{Var } \hat{\theta}_{\text{LS}}$ by v , since $\mathbb{E}[\hat{\theta}_{\text{LS}}] = \theta$, the new **bias** and **variance** are

$$\mathbb{E}[\hat{\theta}_a] - \theta = (a - 1)\theta \quad \text{and} \quad \text{Var } \hat{\theta}_a = a^2 v.$$

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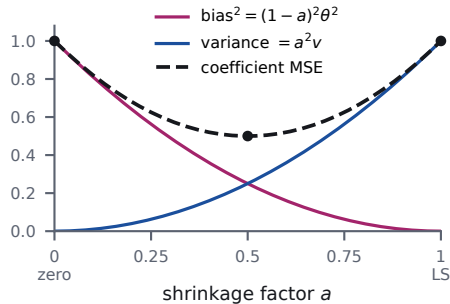
- In this one-coefficient example, ridge does this with $a = \frac{\sum_i x_i^2}{\sum_i x_i^2 + n\lambda}$:

larger λ , smaller a , more shrinkage.

Example: $\theta = 1$ and $v = 1$

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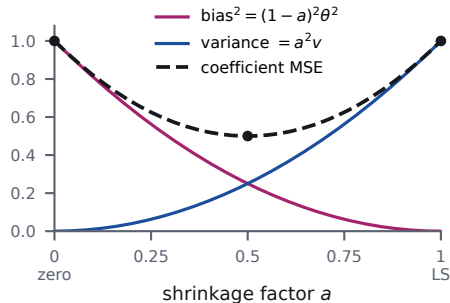
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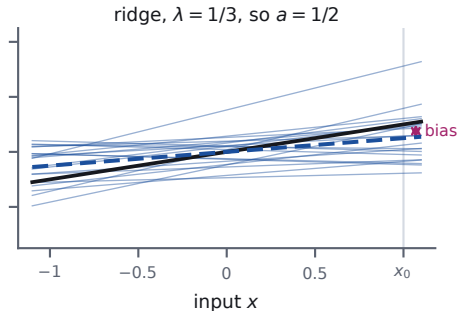
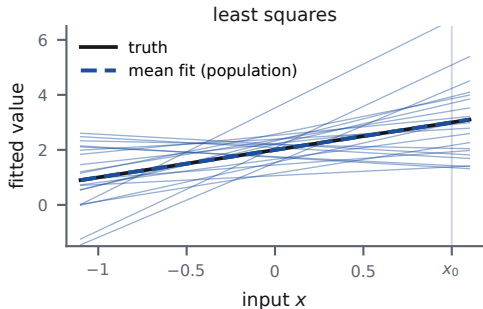
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Note: Shrinking helps when it removes more variance than it adds bias^2 .

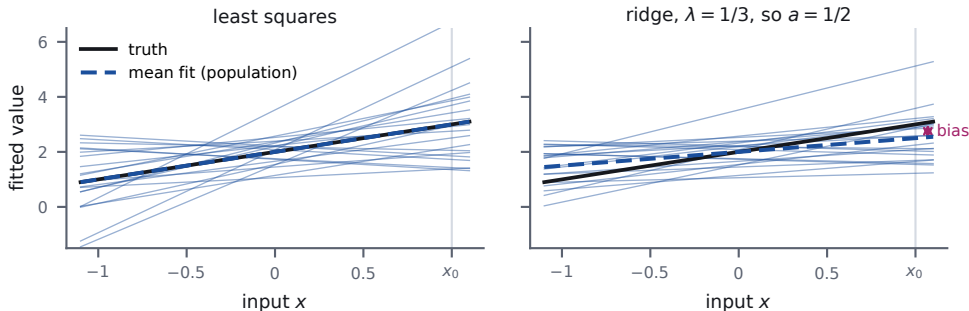
Example: Least squares vs. ridge

Same 20 training sets as before; ridge with $\lambda = 1/3$ held fixed, so $a = 1/2$.



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At $x_0 = 1$: **bias**² + **variance** is $0 + 4/3$ for least squares and $1/4 + 7/12$ for ridge. The slope's variance drops from 1 to $1/4$, the intercept's $1/3$ stays, and both face the same $\sigma^2 = 4$: $\text{MSE}(x_0) = 16/3 \approx 5.33$ vs. $29/6 \approx 4.83$.

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Recall: Maximum likelihood

For the rest of the lecture: θ does *not* include the intercept, and σ^2 is fixed.

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- The **likelihood** is $p(y \mid X, \theta)$.
- Maximum likelihood compares candidate values of θ by how well they explain the observed responses given data X .
- Recall that least squares is the maximum likelihood solution when

$$y \mid X, \theta \sim \mathcal{N}(X\theta, \sigma^2 I_n),$$

specifically

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(y \mid X, \theta) = \arg \min_{\theta} \frac{\|y - X\theta\|^2}{2\sigma^2} = \hat{\theta}_{\text{LS}}.$$

From likelihood to posterior

Idea: Instead of **likelihood** $p(y \mid X, \theta)$, consider **posterior** $p(\theta \mid y, X)$, i.e., given that we've observed X and y , what is the likeliest θ ?

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Definition (Bayes' rule, with θ random and X fixed)

$$p(\theta \mid y, X) = \frac{p(y \mid X, \theta) p(\theta \mid X)}{p(y \mid X)} \quad \text{posterior} \propto \text{likelihood} \times \text{prior}$$

- note that $p(\theta \mid X) = p(\theta)$

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- note that $p(\theta | X) = p(\theta)$
- the **prior** $p(\theta)$ describes what we believe about θ before seeing y
- the denominator is the same constant for every candidate θ :

$$p(y | X) = \int p(y | X, t) p(t) dt.$$

Definition: The MAP estimate

Definition (Maximum a posteriori estimate)

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid y, X)$$

- take negative logarithms:

$$-\log p(\theta \mid y, X) = -\log p(y \mid X, \theta) - \log p(\theta) + \underbrace{\log p(y \mid X)}_{\text{constant in } \theta}$$

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- adding the same constant to every candidate's objective does not change which candidate minimizes it, so

MLE minimizes

$$-\log p(y \mid X, \theta)$$

MAP minimizes

$$-\log p(y \mid X, \theta) - \log p(\theta)$$

Example: MAP for **one** coefficient

One coefficient, with prior $\theta \sim \mathcal{N}(0, \tau^2)$ and

$$\hat{\theta}_{\text{LS}} \mid \theta \sim \mathcal{N}(\theta, v), \quad v = \frac{\sigma^2}{\sum_i x_i^2}.$$

- likelihood $\propto \exp[-(\theta - \hat{\theta}_{\text{LS}})^2/(2v)]$, prior $\propto \exp[-\theta^2/(2\tau^2)]$

Example: MAP for **one** coefficient

One coefficient, with prior $\theta \sim \mathcal{N}(0, \tau^2)$ and

$$\hat{\theta}_{\text{LS}} \mid \theta \sim \mathcal{N}(\theta, v), \quad v = \frac{\sigma^2}{\sum_i x_i^2}.$$

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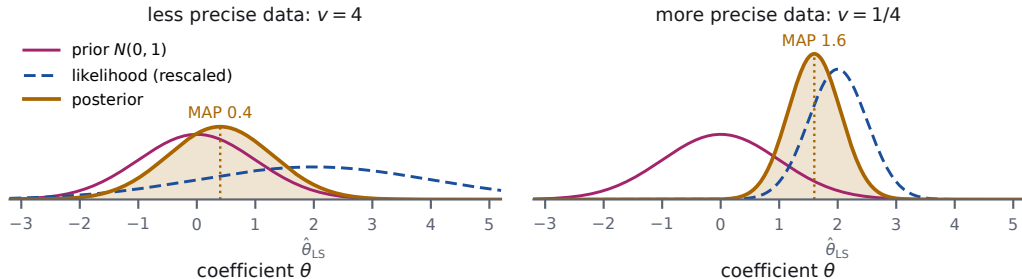
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- a **shrinkage factor** again: $v = \tau^2 = 1$ gives $1/2$, the $a = 1/2$ from before

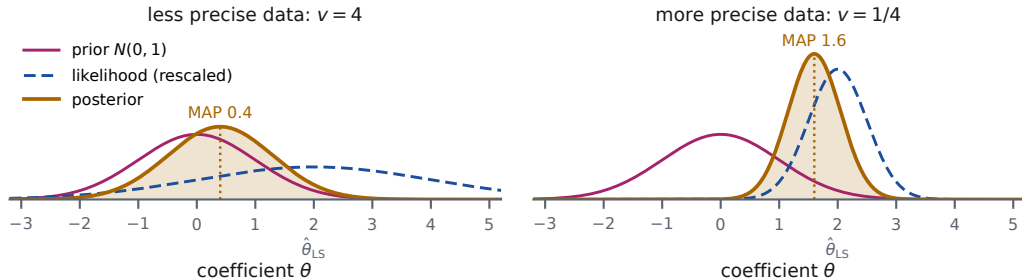
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Key idea: Uncertain estimates are pulled more strongly toward the prior mean; precise data have more influence.

Plan for today

- Bias–variance decomposition 20 min
- Ridge accepts bias to reduce variance 7 min
- From maximum likelihood to MAP 13 min
- MAP with a Gaussian prior 10 min

MAP with a Gaussian prior

Assumptions: θ does not include an intercept; $\sigma^2, \tau^2 > 0$ are fixed constants.

Let y follow the Gaussian linear model and use a Gaussian prior for θ :

$$y \mid X, \theta \sim \mathcal{N}(X\theta, \sigma^2 I_n), \quad \theta \sim \mathcal{N}(0, \tau^2 I_p)$$

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$$-\log p(y \mid X, \theta) = \frac{\|y - X\theta\|^2}{2\sigma^2} + C_1 \quad \text{and} \quad -\log p(\theta) = \frac{\|\theta\|^2}{2\tau^2} + C_2,$$

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We can ignore the constants in the minimization that defines $\hat{\theta}_{\text{MAP}}$, so

$$\hat{\theta}_{\text{MAP}} = \arg \min_{\theta} \{-\log p(y \mid X, \theta) - \log p(\theta)\} = \arg \min_{\theta} \left\{ \frac{\|y - X\theta\|^2}{2\sigma^2} + \frac{\|\theta\|^2}{2\tau^2} \right\}.$$

MAP with Gaussian prior *is* ridge

Multiplying the objective by $2\sigma^2/n > 0$ does not change the minimizer:

$$\hat{\theta}_{\text{MAP}} = \arg \min_{\theta} \left\{ \frac{1}{n} \|y - X\theta\|^2 + \lambda \|\theta\|^2 \right\}, \quad \lambda = \frac{\sigma^2}{n\tau^2}$$

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Watch out: If θ includes an intercept, this prior penalizes it.

Reading λ as a **belief**

$\lambda = \sigma^2 / (n\tau^2)$ is large when

- τ is small: we are confident the coefficients are near 0
- σ is large: the responses are noisy
- n is small: there are few examples, so the prior counts for more

Example: $\sigma^2 = 4$, $n = 12$ and $\tau^2 = 1$ give $\lambda = 4/12 = 1/3$, the penalty in our experiment.

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Remark (Advertising)

If σ^2 were known or estimated separately, the cross-validated $\lambda = 0.1$ would translate into a prior scale $\tau = \sigma / \sqrt{n\lambda}$. Because λ was chosen using the data, this is a data-dependent reading of the penalty.